

Kubernetes

Deployment YAML:

apiVersion: apps/v1

kind: Deployment

metadata:

name: redis-Deployment

namespace: venkatesh-ns

spec:

replicas: 6

selector:

matchlabels:

app: redis

template:

metadata:

labels:

app: redis

spec:

containers:

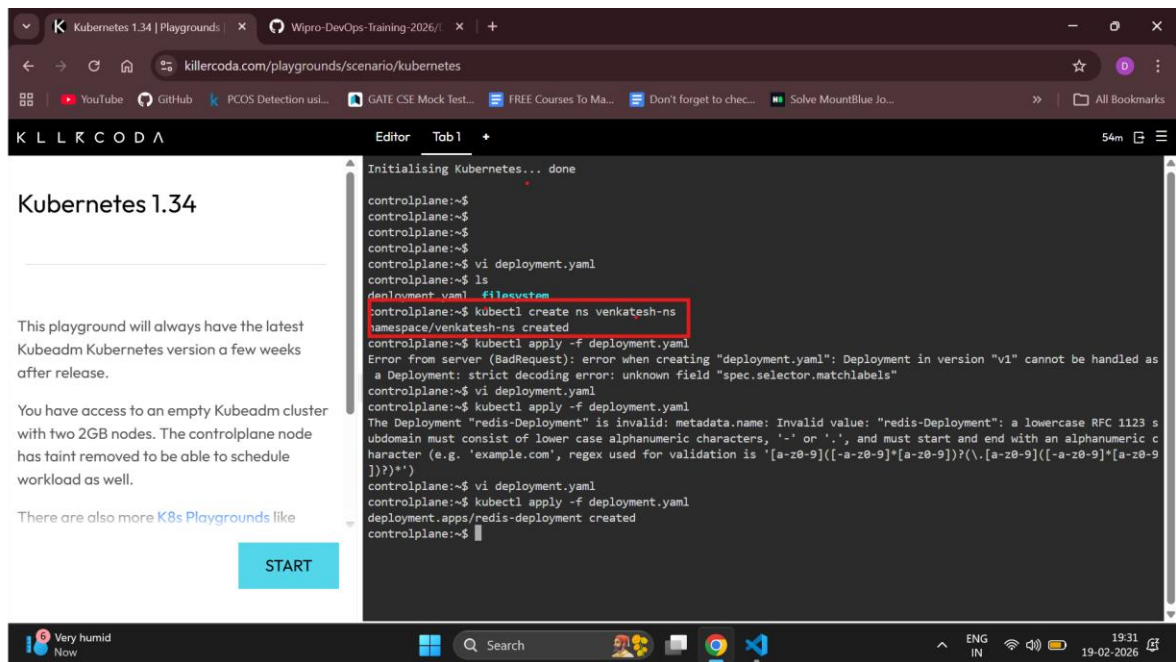
- name: redis-container

image: redis:latest

ports:

- containerPort: 6379

Step-1 Create a namespace



Kubernetes 1.34

This playground will always have the latest Kubeadm Kubernetes version a few weeks after release.

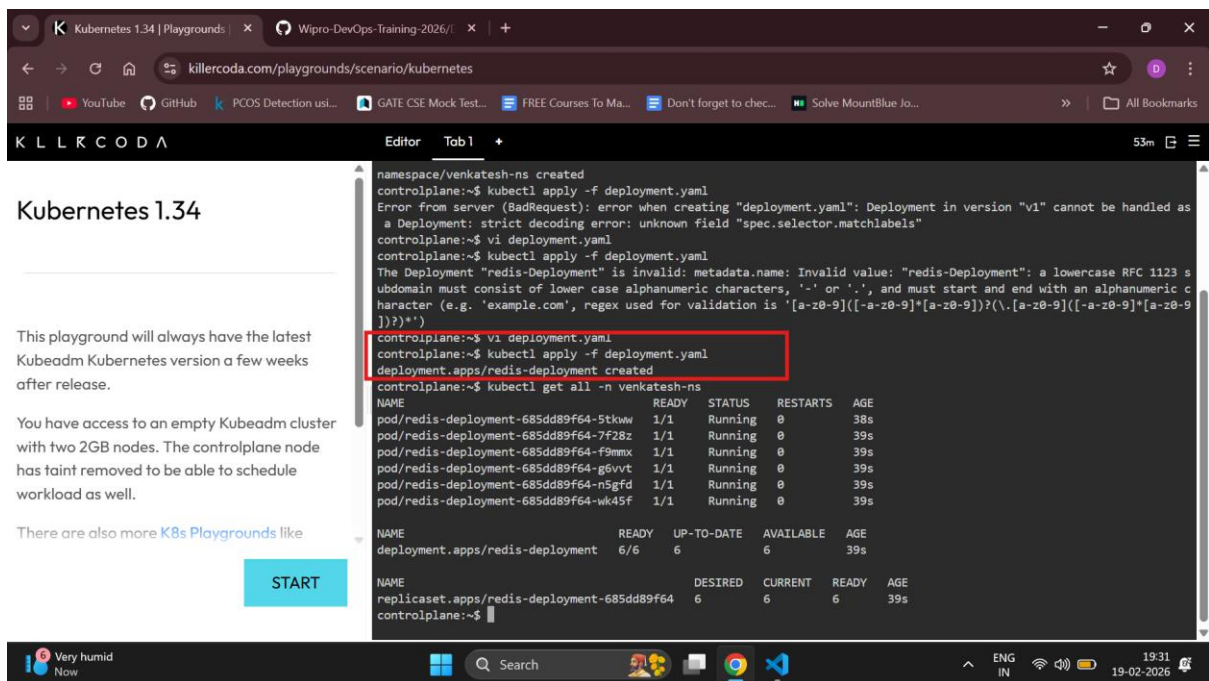
You have access to an empty Kubeadm cluster with two 2GB nodes. The controlplane node has taint removed to be able to schedule workload as well.

There are also more [K8s Playgrounds](#) like

[START](#)

```
Initialising Kubernetes... done
controlplane:~$ 
controlplane:~$ 
controlplane:~$ 
controlplane:~$ 
controlplane:~$ vi deployment.yaml
controlplane:~$ ls
deployment.yaml
controlplane:~$ kubectl create ns venkatesh-ns
namespace/venkatesh-ns created
controlplane:~$ kubectl apply -f deployment.yaml
Error from server (BadRequest): error when creating "deployment.yaml": Deployment in version "v1" cannot be handled as a Deployment: strict decoding error: unknown field "spec.selector.matchLabels"
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
The Deployment "redis-Deployment" is invalid: metadata.name: Invalid value: "redis-Deployment": a lowercase RFC 1123 s
ubdomain must consist of lower case alphanumeric characters, '-' or '.', and must start and end with an alphanumeric c
haracter (e.g. 'example.com', regex used for validation is '[a-z0-9]([-a-z0-9]*[a-z0-9])?(\.[a-z0-9]([-a-z0-9]*[a-z0-9
]))?')
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
deployment.apps/redis-deployment created
controlplane:~$
```

Step 2: Create a deployment yaml file inside a namespace



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[START](#)

```
namespace/venkatesh-ns created
controlplane:~$ kubectl apply -f deployment.yaml
Error from server (BadRequest): error when creating "deployment.yaml": Deployment in version "v1" cannot be handled as
a Deployment: strict decoding error: unknown field "spec.selector.matchLabels"
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
The Deployment "redis-Deployment" is invalid: metadata.name: Invalid value: "redis-Deployment": a lowercase RFC 1123 s
ubdomain must consist of lower case alphanumeric characters, '-' or '.', and must start and end with an alphanumeric c
haracter (e.g. 'example.com', regex used for validation is '[a-z0-9]([-a-z0-9]*[a-z0-9])?(\.[a-z0-9]([-a-z0-9]*[a-z0-9
]))?')
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
deployment.apps/redis-deployment created
controlplane:~$ kubectl get all -n venkatesh-ns
NAME                                READY    STATUS    RESTARTS   AGE
pod/redis-deployment-685dd89f64-5tkw 1/1      Running   0           38s
pod/redis-deployment-685dd89f64-7f28z 1/1      Running   0           39s
pod/redis-deployment-685dd89f64-f9mmx 1/1      Running   0           39s
pod/redis-deployment-685dd89f64-g6vvt 1/1      Running   0           39s
pod/redis-deployment-685dd89f64-n5gfd 1/1      Running   0           39s
pod/redis-deployment-685dd89f64-wk45f 1/1      Running   0           39s
NAME                                READY    UP-TO-DATE   AVAILABLE   AGE
deployment.apps/redis-deployment     6/6      6             6           39s
NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/redis-deployment-685dd89f64 6         6         6       39s
controlplane:~$
```

Step 3: After creating YAML file u need to check pods and replicas and deployments

The screenshot shows a web browser window with the URL `killercoda.com/playgrounds/scenario/kubernetes`. The page title is "Kubernetes 1.34". On the left, there is a sidebar with the text "Kubernetes 1.34" and a "START" button. The main area is a terminal window showing the following commands and output:

```
namespace/venkatesh-ns created
controlplane:~$ kubectl apply -f deployment.yaml
Error from server (BadRequest): error when creating "deployment.yaml": Deployment in version "v1" cannot be handled as a Deployment: strict decoding error: unknown field "spec.selector.matchLabels"
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
The Deployment "redis-deployment" is invalid: metadata.name: Invalid value: "redis-deployment": a lowercase RFC 1123 subdomain must consist of lower case alphanumeric characters, '-' or '.', and must start and end with an alphanumeric character (e.g. 'example.com', regex used for validation is '[a-z0-9]([-a-z0-9]*[a-z0-9])?(\.[a-z0-9]([-a-z0-9]*[a-z0-9])?)*')
controlplane:~$ vi deployment.yaml
controlplane:~$ kubectl apply -f deployment.yaml
deployment.apps/redis-deployment created
controlplane:~$ kubectl get all -n venkatesh-ns
```

NAME	READY	STATUS	RESTARTS	AGE
pod/redis-deployment-685dd89f64-5tkww	1/1	Running	0	38s
pod/redis-deployment-685dd89f64-7f28z	1/1	Running	0	39s
pod/redis-deployment-685dd89f64-f9mxx	1/1	Running	0	39s
pod/redis-deployment-685dd89f64-g6vvt	1/1	Running	0	39s
pod/redis-deployment-685dd89f64-n5gfd	1/1	Running	0	39s
pod/redis-deployment-685dd89f64-wk45f	1/1	Running	0	39s

```
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/redis-deployment    6/6      6              6             39s
```

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/redis-deployment-685dd89f64	6	6	6	39s

Step 4: After checking pods we need to describe the pod

The screenshot shows the same web browser window as Step 3. The terminal window now shows the following commands and output:

```
controlplane:~$ kubectl describe pod redis-deployment-685dd89f64-5tkww -n venkatesh-ns
```

The output of the `kubectl describe` command is shown below:

```
Name:         redis-deployment-685dd89f64-5tkww
Namespace:    venkatesh-ns
Priority:      0
Service Account: default
Node:         controlplane/172.30.1.2
Start Time:   Thu, 19 Feb 2026 14:08:54 +0000
Labels:       pod-template-hash=685dd89f64
Annotations:  cni.projectcalico.org/containerID: a0e66a72a5489ef338c53939801d16a33cc3af65e31df7c479cd1a3893b25b68
              cni.projectcalico.org/podIP: 192.168.0.6/32
              cni.projectcalico.org/podIPs: 192.168.0.6/32
Status:       Running
IP:           192.168.0.6
IPs:          IP: 192.168.0.6
Controlled By: ReplicaSet/redis-deployment-685dd89f64
Containers:
  redis-container:
    Container ID:  containerd://41cba2953831b33350906c35d7f044abb4ddeb612e1ed69395fdebc7f2dfd8af
    Image:         redis:latest
    Image ID:      docker.io/library/redis@sha256:7b6fb55d8b0adcd77269dc52b3cffffe5f59ca5d43dec3c9db0e18aacce7942e1
    Port:          6379/TCP
    Host Port:     0/TCP
    State:         Running
```

Step 5: Again, u have to upgrade the version by changing the image name in the yaml file

Kubernetes 1.34

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There are also more [K8s Playgrounds](#) like

START

```

Normal Started 103s kubelet Started container redis-container
Error from server (NotFound): pods "redis-deployment" not found
controlplane:~$ vi deployment.yaml
controlplane:~$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: redis-deployment
  namespace: venkatesh-ns
spec:
  replicas: 6
  selector:
    matchLabels:
      app: redis
  template:
    metadata:
      labels:
        app: redis
    spec:
      containers:
        - name: redis-container
          image: redis:8.2.4-alpine
          ports:
            - containerPort: 6379

controlplane:~$ kubectl apply -f deployment.yaml
deployment.apps/redis-deployment configured
controlplane:~$

```

Step 6: After change the image name in the yaml file. To check the pods and replica set and deployment.

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START

```

metadata:
  labels:
    app: redis
spec:
  containers:
    - name: redis-container
      image: redis:8.2.4-alpine
      ports:
        - containerPort: 6379

controlplane:~$ kubectl apply -f deployment.yaml
deployment.apps/redis-deployment configured
controlplane:~$ kubectl get all -n venkatesh-ns
NAME                                READY    STATUS    RESTARTS   AGE
pod/redis-deployment-7854664c67-9pf2g 1/1      Running   0           27s
pod/redis-deployment-7854664c67-dnrjb 1/1      Running   0           26s
pod/redis-deployment-7854664c67-g8pxx 1/1      Running   0           35s
pod/redis-deployment-7854664c67-tkhdv 1/1      Running   0           28s
pod/redis-deployment-7854664c67-tvd5h 1/1      Running   0           34s
pod/redis-deployment-7854664c67-tvfkx 1/1      Running   0           35s

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/redis-deployment    6/6      6              6            5m6s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/redis-deployment-685dd89f64 0          0          0        5m6s
replicaset.apps/redis-deployment-7854664c67 6          6          6        35s
controlplane:~$

```

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START

```

cni.projectcalico.org/podIP: 192.168.0.8/32
cni.projectcalico.org/podIPs: 192.168.0.8/32
Status:
IP:
192.168.0.8
IPs:
192.168.0.8
Controlled By: ReplicaSet/redis-deployment-7854664c67
Containers:
redis-container:
Container ID: containerd://db0e95efbcc920fd2d6fd67c2fd3aff8b752766bb361c9f6282616697e4558e4
Image: redis:8.2.4-alpine
Image ID: docker.io/library/redis@sha256:a308ca111032fa8f306a2dc7be7ba5deb8b777ed5d258c733cddb48a1fd7904
Port: 6379/TCP
Host Port: 0/TCP
State: Running
Started: Thu, 19 Feb 2026 14:05:40 +0000
Ready: True
Restart Count: 0
Environment: <none>
Mounts:
/var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-pxbb9 (ro)
Conditions:
Type Status
PodReadyToStartContainers True
Initialized True
Ready True
ContainersReady True
PodScheduled True

```

Step 7: To roll back the image name using command and check the history of deployment

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START

```

g waiting). Image size: 27559171 bytes.
Normal Created 87s kubelet Created container: redis-container
Normal Started 87s kubelet Started container redis-container
Error from server (NotFound): pods "redis-deployment" not found
controlplane:~$ kubectl get all -n venkatesh-ns
NAME READY STATUS RESTARTS AGE
pod/redis-deployment-7854664c67-9pf2g 1/1 Running 0 2m15s
pod/redis-deployment-7854664c67-dnrjb 1/1 Running 0 2m14s
pod/redis-deployment-7854664c67-g8pxx 1/1 Running 0 2m23s
pod/redis-deployment-7854664c67-tkhdv 1/1 Running 0 2m8s
pod/redis-deployment-7854664c67-tvd5h 1/1 Running 0 2m22s
pod/redis-deployment-7854664c67-tvfkn 1/1 Running 0 2m23s
NAME READY UP-TO-DATE AVAILABLE AGE
deployment.apps/redis-deployment 6/6 6 6 6m54s
NAME DESIRED CURRENT READY AGE
replicaset.apps/redis-deployment-685dd89f64 0 0 0 6m54s
replicaset.apps/redis-deployment-7854664c67 6 6 6 2m23s
controlplane:~$ kubectl rollout undo deployment redis-deployment -n venkatesh-ns
deployment.apps/redis-deployment rolled back
controlplane:~$ kubectl rollout history deployment redis-deployment -n venkatesh-ns
deployment.apps/redis-deployment
REVISION CHANGE-CAUSE
2 <none>
3 <none>
controlplane:~$

```

Step 8: Finally, u check the history of the deployment file

Kubernetes 1.34

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There are also more [K8s Playgrounds](#) like

START

```

replicaset.apps/redis-deployment-685dd89f64 0 0 0 6m54s
replicaset.apps/redis-deployment-7854664c67 6 6 6 2m23s
controlplane:~$ kubectl rollout undo deployment redis-deployment -n venkatesh-ns
deployment.apps/redis-deployment rolled back
controlplane:~$ kubectl rollout history deployment redis-deployment -n venkatesh-ns
deployment.apps/redis-deployment
REVISION  CHANGE-CAUSE
2         <none>
3         <none>

controlplane:~$ kubectl rollout status deployment redis-deployment -n venkatesh-ns
deployment "redis-deployment" successfully rolled out
controlplane:~$ kubectl get all -n venkatesh-ns
NAME                                READY    STATUS    RESTARTS   AGE
pod/redis-deployment-685dd89f64-2vvp 1/1      Running   0          77s
pod/redis-deployment-685dd89f64-5vx8 1/1      Running   0          80s
pod/redis-deployment-685dd89f64-h7vr 1/1      Running   0          80s
pod/redis-deployment-685dd89f64-jgxb 1/1      Running   0          79s
pod/redis-deployment-685dd89f64-jkmc 1/1      Running   0          76s
pod/redis-deployment-685dd89f64-15fb 1/1      Running   0          75s

NAME                                READY    UP-TO-DATE   AVAILABLE   AGE
deployment.apps/redis-deployment 6/6      6            6           9m10s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/redis-deployment-685dd89f64 6         6         6       9m10s
replicaset.apps/redis-deployment-7854664c67 0         0         0       4m39s
controlplane:~$
  
```

Step 9: After rollback the image we need to describe the pod .

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There are also more [K8s Playgrounds](#) like

START

```

Name:         venkatesh-ns
Priority:      0
Service Account: default
Node:         controlplane/172.30.1.2
Start Time:   Thu, 19 Feb 2026 14:08:46 +0000
Labels:       app=redis
              pod-template-hash=685dd89f64
Annotations:  cnf.projectcalico.org/containerID: 5635b473c8b337f753c61afe85bdeca8246da89df3860d276333c32439f6c8b1
              cnf.projectcalico.org/podIP: 192.168.0.11/32
              cnf.projectcalico.org/podIPs: 192.168.0.11/32
Status:       Running
IP:           192.168.0.11
IPs:          192.168.0.11
Controlled By: ReplicaSet/redis-deployment-685dd89f64
Containers:
  redis-container:
    Container ID:   containerd://5d8d47530b8232243aaac3f542b1d504369c978ead17e95f2f9beb8e10445584
    Image:          redis:latest
    Image ID:       docker.io/library/redis@sha256:7b6fb55d8b0adcd77269dc52b3cffff5f59ca5d43dec3c90dbe18aacce7942e1
    Port:          6379/TCP
    Host Port:     0/TCP
    State:          Running
      Started:      Thu, 19 Feb 2026 14:08:50 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
  
```